Research Evidence

Research Gap: Existing explainability methods such as SHAP and LIME are widely used but fail to capture temporal dependencies in time series data. Attention-based approaches provide some interpretability but lack reliability and consistency. Additionally, there is no standard framework to evaluate explanation quality in temporal settings. This highlights a gap in developing temporally-aware and clinically meaningful explanation methods for ICU prediction models.

CRAAP Analysis for Major Papers

Paper	Currency	Relevance	Authority	Accuracy	Purpose	Key Insight
Lundberg & Lee (2017) – SHAP	Foundational (still widely used)	High – baseline explainability	NeurIPS (top-tier)	Strong theoretical grounding	Explain model predictions	Produces feature importance but ignores temporal dependencies
Ribeiro et al. (2016) – LIME	Foundational	High – widely used baseline	KDD (top-tier)	Locally accurate but unstable	Interpret black-box models	Treats time series as static data
Lim et al. (2021) – TFT	Recent	Very high – healthcare time series	Google Research	Robust deep learning model	Combine performance + interpretability	Attention used, but not fully reliable as explanation
Jain & Wallace (2019)	Recent	High – critiques attention	NAACL	Strong empirical evidence	Test interpretability claims	Attention weights do not reflect true model reasoning
Doshi-Velez & Kim (2017)	Foundational	High – evaluation framework	Harvard/MIT researchers	Conceptually strong	Define interpretability evaluation	Lack of standard evaluation metrics
Tschernutter et al. (2023)	Very recent	Very high – directly relevant	Recent XAI research	Comprehensive survey	Review time series explainability	Highlights lack of temporal-aware explanation methods

These papers were selected as foundational sources based on their influence, relevance to time series explainability, and their role in identifying key limitations that motivate the proposed research gap.

Paper	Model Used	Explainability Method	Limitation
SHAP (Lundberg & Lee, 2017)	Model-agnostic	SHAP values	No temporal awareness
LIME (Ribeiro et al., 2016)	Model-agnostic	Local explanations	Unstable and static interpretation
TFT (Lim et al., 2021)	Transformer	Attention-based	Attention not fully reliable
LSTM Models (Lipton et al., 2015)	Sequential model	None	Black-box nature
Jain & Wallace (2019)	Various	Attention analysis	Attention not faithful
Time Series XAI (Tschernutter et al., 2023)	Various	Survey	No unified framework